

Formulary

Real, Imaginary, and Complex Numbers

RATIONAL EXPONENTS

$$x^{\frac{a}{b}} = \sqrt[b]{x^a}$$

POWERS OF i

$$i = \sqrt{-1}$$

$$i^2 = -1$$

$$i^3 = -i$$

$$i^4 = 1$$

ADDING COMPLEX NUMBERS

For any real a , b , c , and d , if $w = a + bi$ and $z = c + di$, then $w + z = (a + c) + (b + d)i$.

Factoring Patterns

PERFECT SQUARE TRINOMIAL PATTERNS

$$a^2 + 2ab + b^2 = (a + b)^2$$

DIFFERENCE OF SQUARES PATTERNS

$$a^2 - b^2 = (a + b)(a - b)$$

SUM OF CUBES

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

DIFFERENCE OF CUBES

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Plane Geometry

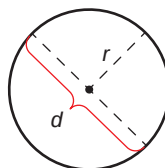
CIRCLE

circumference $C = \pi d = 2\pi r$

area $A = \pi r^2$

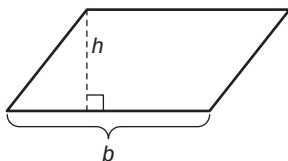
length of an arc with degree measure m $L = \left(\frac{m}{360^\circ}\right)2\pi r$

equation with center (h, k) on a coordinate plane $(x - h)^2 + (y - k)^2 = r^2$



PARALLOGRAM

area $A = bh$



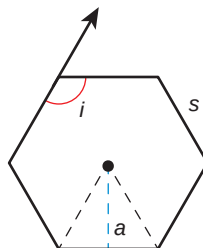
REGULAR POLYGON WITH n SIDES

sum of interior angles $I = (n - 2)180^\circ$

interior angle of regular polygon $i = \frac{(n - 2)180^\circ}{n}$

perimeter of regular polygon $P = ns$

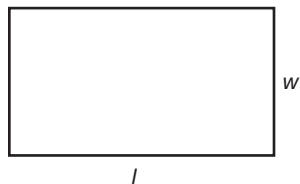
area of regular polygon $A = \frac{1}{2}aP$



RECTANGLE

area $A = lw$

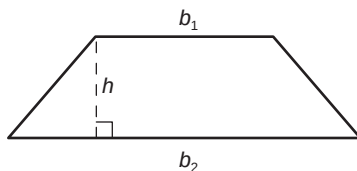
perimeter $P = 2l + 2w$



TRAPEZOID

length of midsegment $\text{length} = \frac{b_1 + b_2}{2}$

area $A = \frac{1}{2}h(b_1 + b_2)$

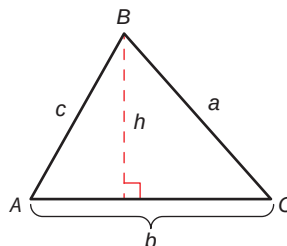


TRIANGLE: GENERAL

sum of interior angles $m\angle A + m\angle B + m\angle C = 180^\circ$

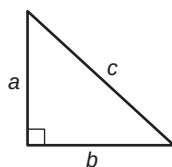
area $A = \frac{1}{2}bh$

length of midsegment $\text{length} = \frac{1}{2}$ length of parallel side



TRIANGLE: RIGHT

Pythagorean theorem $a^2 + b^2 = c^2$

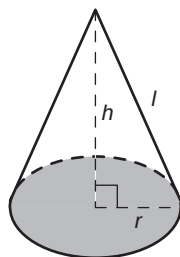


Solid Geometry

CONE

volume $V = \frac{1}{3}Bh = \frac{1}{3}\pi r^2 h$

surface area $S = L + B = \pi r l + \pi r^2$

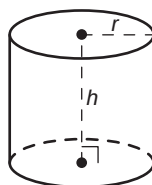


CYLINDER

volume $V = Bh = \pi r^2 h$

surface area $S = L + 2B$

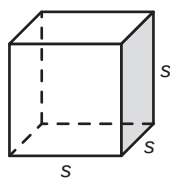
$$S = 2\pi r h + 2\pi r^2$$



PRISM: CUBE

volume $V = s^3$

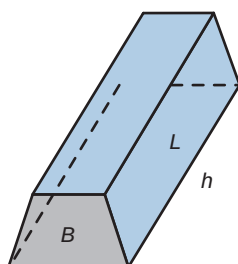
surface area $S = 6s^2$



PRISM: GENERAL

volume $V = Bh$

surface area $S = 2B + L$

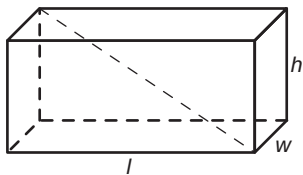


PRISM: RIGHT RECTANGULAR

volume $V = lwh$

surface area $S = 2lw + 2lh + 2wh$

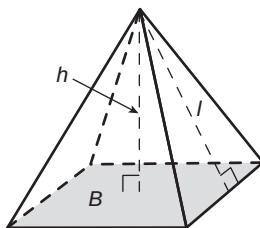
length of diagonal $d = \sqrt{l^2 + w^2 + h^2}$



PYRAMID

volume $V = \frac{1}{3}Bh$

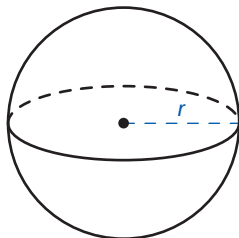
surface area $S = \frac{1}{2}lP + B$



SPHERE

volume $V = \frac{4}{3}\pi r^3$

surface area $S = 4\pi r^2$



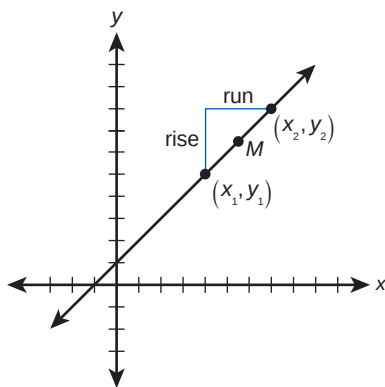
Coordinate Geometry

LINE AND SEGMENT

slope $m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$

coordinates of midpoint $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

distance $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$



LINEAR EQUATION FORMS

standard $Ax + By = C$

slope-intercept $y = mx + b$

point-slope $y - y_1 = m(x - x_1)$

CIRCLE

equation in graphing form $(x - h)^2 + (y - k)^2 = r^2$

center (h, k) and radius r

PARABOLA GRAPHING FORMS

vertical axis of symmetry

$$y - k = a(x - h)^2 \text{ (axis of symmetry } x = h)$$

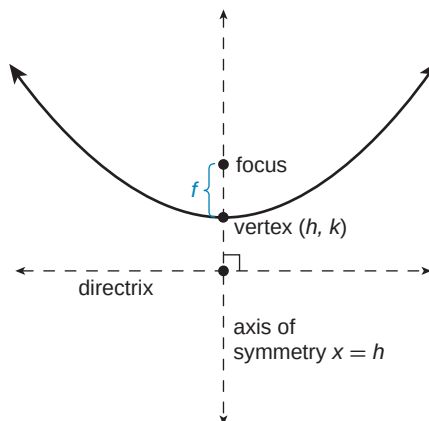
horizontal axis of symmetry

$$x - h = a(y - k)^2 \text{ (axis of symmetry } y = k)$$

vertex (h, k)

focal distance $f = \frac{1}{4a}$

eccentricity $e = 1$

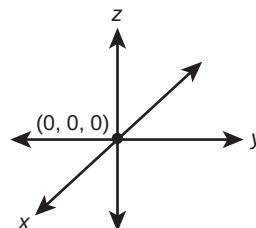


THREE-DIMENSIONAL SPACE

distance between points $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$

coordinates of midpoint $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right)$

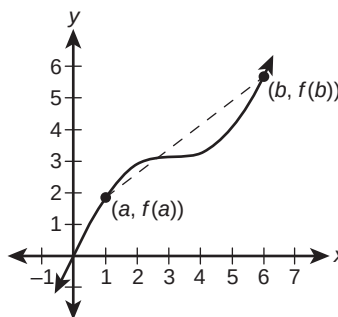
general equation of a line $Ax + By + Cz = D$



FUNCTIONS

average rate of change $\frac{f(b) - f(a)}{b - a}$

axis of symmetry for $f(x) = ax^2 + bx + c$ $x = -\frac{b}{2a}$



Solving Equations

SOLVING QUADRATICS

completing the square Given the equation $ax^2 + bx = c$, add $\left(\frac{b}{2a}\right)^2$ to both sides.

discriminant Given the equation $ax^2 + bx + c = 0$, the discriminant is $b^2 - 4ac$.

quadratic formula The solutions of the equation $ax^2 + bx + c = 0$, where $a \neq 0$,

are $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

EXPONENTS AND LOGARITHMS

$$x^a \cdot x^b = x^{a+b}$$

$$\frac{x^a}{x^b} = x^{a-b}$$

$$x^0 = 1$$

$$x^1 = x$$

For any $a > 0$ and $b > 0$, $b \neq 1$: $\log_b a = x$ if and only if $b^x = a$.

$$\log mn = \log m + \log n$$

$$\log\left(\frac{m}{n}\right) = \log m - \log n$$

$$\log m^a = a \log m$$

Counting and Probability

FACTORIAL

$$n! = n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot 1 \text{ (} n \text{ factors)}$$

$$0! = 1$$

SIMPLE THEORETICAL PROBABILITY

$$P(E) = \frac{\text{number of outcomes in event } E}{\text{total number of outcomes in sample space } S} = \frac{n(E)}{n(S)}$$

PROBABILITY OF DEPENDENT EVENTS

$$P(A \text{ and } B) = P(A) \cdot P(B|A)$$

PROBABILITY OF INDEPENDENT EVENTS

$$P(A \text{ and } B) = P(A) \cdot P(B)$$

PROBABILITY OF MUTUALLY EXCLUSIVE EVENTS

$$P(A \text{ or } B) = P(A) + P(B)$$

PROBABILITY OF COMPLEMENTARY EVENTS

$$P(A) = 1 - P(B)$$

EXPERIMENTAL PROBABILITY OF EVENT E

$$P(E) = \frac{\text{number of times event } E \text{ has occurred}}{n}$$

Statistics

MEAN

For a data set with n elements, the mean is

$$\bar{x} = \frac{x_1 + x_1 + \dots + x_n}{n}.$$

MEDIAN

Arrange the values in order from least to greatest. For an

Odd number of values, use the middle value.

Even number of values, use the average of the middle two values.

MODE

The mode is the value that occurs most often in a set of data. If no one value occurs most often, then there is no mode for the set.

STANDARD DEVIATION

To find the standard deviation s of a data set with n values, where x is the data value and $\bar{x}$ is the mean, use the formula

$$s = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n - 1}}.$$

z-SCORE

If x is a raw data value from a normally distributed data set with mean μ and standard deviation σ , then

$$z = \frac{x - \mu}{\sigma}$$

is the number of standard deviations x is from the mean.

SAMPLING STANDARD DEVIATION

If random samples of size n are taken from a distribution with population standard deviation σ , then the sampling distribution will have standard deviation approximately equal to

$$\frac{\sigma}{\sqrt{n}}.$$

STANDARD DEVIATION OF A SAMPLING DISTRIBUTION OF A PROPORTION

If a sample of size n is drawn from a population with proportion p , then the sampling distribution of the proportion will have standard deviation approximately equal to

$$\sqrt{\frac{p(1 - p)}{n}}.$$

Sequences and Series

SEQUENCES: ARITHMETIC

common difference of an arithmetic sequence $d = a_n - a_{n-1}$

iterative rule for an arithmetic sequence $a_n = a_1 + (n-1)d$

recursive rule for an arithmetic sequence $a_n = a_{n-1} + d$

SEQUENCES: GEOMETRIC

common ratio of a geometric sequence $r = \frac{a_n}{a_{n-1}}$

iterative rule for a geometric sequence $a_n = a_1 \cdot r^{n-1}$

recursive rule for a geometric sequence $a_n = r \cdot a_{n-1}$

GENERAL FORMULA FOR THE SUM OF A SERIES

sigma notation The sum of the first n terms of a sequence can be written as

$$S_n = \sum_{i=1}^n a_i = a_1 + a_2 + a_3 + \dots + a_n$$

where i is the index, 1 is the lower limit, and n is the upper limit.

arithmetic series The n th partial sum of an arithmetic series a with common difference d is

$$S_n = \frac{n}{2}(a_1 + a_n) \text{ or } S_n = \frac{n}{2}(2a_1 + (n-1)d).$$

geometric series The n th partial sum of a geometric series a with common ratio r is

$$S_n = \frac{a_1(1-r^n)}{1-r}, \text{ or } S_n = \frac{a_1 - a_n r}{1-r} \text{ where } r \neq 1.$$

Exponential Growth and Decay

EXPONENTIAL DECAY FORMULA

If a quantity is decaying exponentially from initial amount b and with decay rate r , then the amount y remaining after t time periods is

$$y = b(1-r)^t.$$

EXPONENTIAL GROWTH FORMULA

If a quantity is growing exponentially from the initial amount b where r is the fixed percent expressed as a decimal, then the total amount y after t time periods is

$$y = b(1 + r)^t.$$

COMPOUND INTEREST FORMULA

The total amount A of an investment with initial principal P , earning compound interest at an annual interest rate r and compounded n times per year for t years, is

$$A = P \left(1 + \frac{r}{n} \right)^{nt}.$$

HALF-LIFE FORMULA

The amount y of a radioactive substance after t time periods, where b is the initial amount and h is the half-life, is

$$y = b \left(\frac{1}{2} \right)^{\frac{t}{h}}.$$

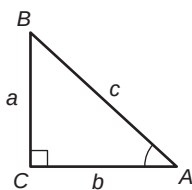
Trigonometry

RIGHT TRIANGLE RATIOS

tangent: $\tan A = \frac{\text{opposite}}{\text{adjacent}} = \frac{a}{b}$

cosine: $\cos A = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{b}{c}$

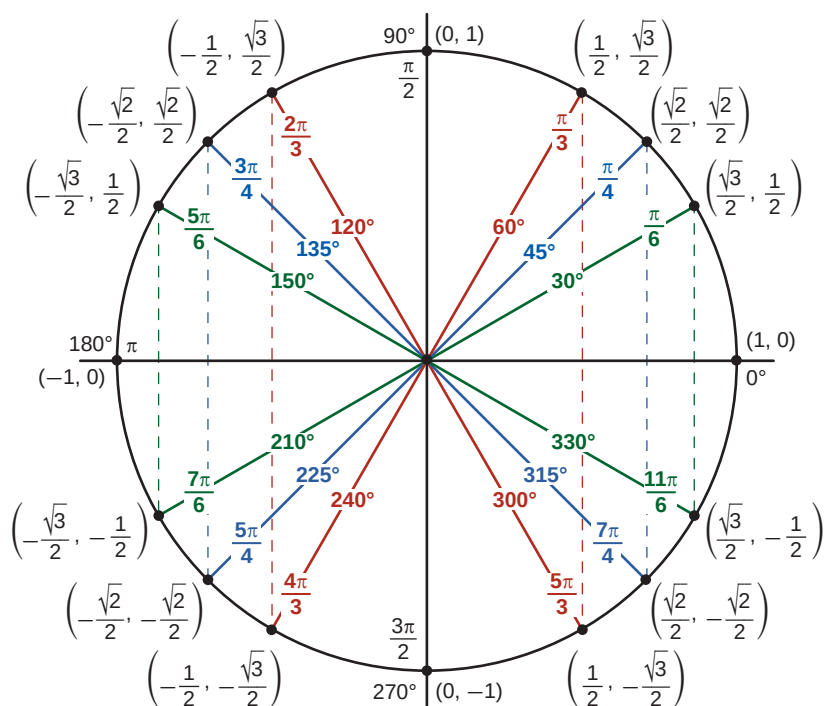
sine: $\sin A = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{a}{c}$



SPECIAL RIGHT TRIANGLES

Angle measure	Sine	Cosine	Tangent
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
45°	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$

UNIT CIRCLE ON THE COORDINATE PLANE



TRIGONOMETRIC IDENTITIES

$$\sin^2 x + \cos^2 x = 1$$

$$\tan x = \frac{\sin x}{\cos x}$$

$$\sin x = \cos(90^\circ - x)$$

$$\cos x = \sin(90^\circ - x)$$

General Applications

DISTANCE

For uniform motion, where d is distance, r is rate, and t is time,

$$d = rt.$$

PROJECTILE MOTION

The height of an object, in meters, after t seconds, with initial vertical velocity v_0 and initial height h_0 , is given by

$$h(t) = -4.9t^2 + v_0t + h_0.$$

The height of an object, in feet, after t seconds, with initial vertical velocity v_0 and initial height h_0 , is given by

$$h(t) = -16t^2 + v_0t + h_0.$$

SIMPLE INTEREST

The amount of simple interest I earned, where P is the principal (amount borrowed, deposited, or invested), r is the annual interest rate, and t is the time in years, is given by

$$I = Prt.$$

TEMPERATURE CONVERSION

$$F = \frac{9}{5}C + 32$$

where F is degrees Fahrenheit and C is degrees Celsius.